India Breathes Bad Air

India has a National Air Quality Index (AQI) since 2014 that takes eight pollutants into account: particulate matter less than 10 micrometre (PM_{10}); particulate matter less than 2.5 micrometre ($PM_{2.5}$); nitrogen dioxide (NO_2); sulphur dioxide (SO_2); carbon monoxide (CO); ozone (O_3); ammonia (NH_3); and lead (Pb). Their individual values over a twenty-four-hour or eight-hour period are placed in six categories depending upon safety levels: good (0–50), satisfactory (51–100), moderately polluted (101–200), poor (201–300), very poor (301–400) and severe (401–500). The most severe value of any of these eight elements indicates the AQI for that place and day.

Particulate matter, especially $PM_{2.5}$, enters the lungs and can cause aggravated asthma, decreased lung function, respiratory difficulties like irritation of airways, coughing and difficulty in breathing, and even premature death in people with heart or lung disease. NO_2 also contributes to particulate pollution and chemical reactions that produce ozone. Its impact on health is largely similar to the effects of PM. SO_2, besides causing health problems like NO_2, is also linked to cardiovascular disease. CO causes headache, dizziness, vomiting and nausea; if the levels are very high, it may lead to unconsciousness and even death. Ozone can trigger chest pain and coughing and harm lung tissues, besides worsening bronchitis and asthma. A large presence of ammonia in the air causes burning of the nose, throat and respiratory tract. It can also cause bronchiolar and alveolar oedema. Exposure to lead causes anaemia and may damage the kidneys and brain.

Air is life-giving and life-sustaining. However, the AQI readings tell us how injurious life-giving air has become to life itself. ‘Delhi’s AQI Reads 450 and Reaches Most Severe Zone’—we’ve all seen screaming newspaper